Xelulu — Wiring Tables

Brain = **ESP32-S3 (N16R8)**. No external microSD module — using the board's built-in SD slot (needs no wires).

Read each row as: **this pin on Module A → that pin on Module B**. "5V", "3.3V", "GND" = the power rails from the Power table (§1).

> Build order: wire §1 Power, check 5.0 V and 3.3 V with a multimeter, **then** connect the ESP, then one module at a time.

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1. Power

Module A	Pin A	→ Module B	Pin B	Note
18650 cell 1	−	18650 cell 2	+	cells in **series** (2S)
18650 pack	cell-2 −	2S BMS	B−	
18650 pack	cell-1 − / cell-2 + junction	2S BMS	BM	middle tap
18650 pack	cell-1 +	2S BMS	B+	
2S BMS	P+	KCD1 switch	pin 1	
KCD1 switch	pin 2	XY-3606 buck	IN+	= **7.4V** line
2S BMS	P−	XY-3606 buck	IN−	= **GND**
XY-3606 buck	OUT+	**5V rail**	—	set pot to 5.0 V **before** loading
XY-3606 buck	OUT−	**GND rail**	—	
5V rail	—	AMS1117	IN	
GND rail	—	AMS1117	GND	
AMS1117	OUT	**3.3V rail**	—	
Type-C breakout	VBUS	1794301 charger	IN+	
Type-C breakout	GND	1794301 charger	IN−	
1794301 charger	B+	2S BMS	B+	charging
1794301 charger	B−	2S BMS	B−	charging

All GNDs connect together (one common ground).

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2. ESP32-S3 power

Module A	Pin A	→ Module B	Pin B
ESP32-S3	5V	5V rail	—
ESP32-S3	GND	GND rail	—

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3. Face display — 2" IPS ST7789 (SPI)

Module A	Pin A	→ Module B	Pin B
Display	VCC	3.3V rail	—
Display	GND	GND rail	—
Display	SCL	ESP32-S3	GPIO12
Display	SDA	ESP32-S3	GPIO11
Display	RST	ESP32-S3	GPIO21
Display	DC	ESP32-S3	GPIO14
Display	CS	ESP32-S3	GPIO10
Display	EN	ESP32-S3	GPIO45
Display	TE	(not connected)	—

EN = backlight on: the firmware sets GPIO45 HIGH about 0.5 s after power-up (as `Display_test_code` does). GPIO45 is a boot strapping pin and must be LOW at reset, so don't add a pull-up on EN. TE is an optional sync output — not needed.

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4. Microphones — 2× GY-SPH0645 (I2S)

Module A	Pin A	→ Module B	Pin B
Mic LEFT	3V	3.3V rail	—
Mic LEFT	GND	GND rail	—
Mic LEFT	BCLK	ESP32-S3	GPIO4
Mic LEFT	LRCL	ESP32-S3	GPIO5
Mic LEFT	DOUT	ESP32-S3	GPIO6
Mic LEFT	SEL	GND rail	—
Mic RIGHT	3V	3.3V rail	—

Module A	Pin A	→ Module B	Pin B
Mic RIGHT	GND	GND rail	—
Mic RIGHT	BCLK	ESP32-S3	GPIO4
Mic RIGHT	LRCL	ESP32-S3	GPIO5
Mic RIGHT	DOUT	ESP32-S3	GPIO6
Mic RIGHT	SEL	3.3V rail	—

Both mics share GPIO4/5/6. SEL is the only difference (GND = left, 3.3V = right).

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5. Speaker amp — MAX98357 (I2S)

Module A	Pin A	→ Module B	Pin B
MAX98357	VIN	5V rail	—
MAX98357	GND	GND rail	—
MAX98357	BCLK	ESP32-S3	GPIO15
MAX98357	LRC	ESP32-S3	GPIO16
MAX98357	DIN	ESP32-S3	GPIO7
MAX98357	SD	3.3V rail	—
MAX98357	GAIN	(not connected)	—
MAX98357	+	Speaker 8Ω	+
MAX98357	–	Speaker 8Ω	–

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6. Touch — MPR121 (I2C)

Module A	Pin A	→ Module B	Pin B
MPR121	VIN	3.3V rail	—
MPR121	GND	GND rail	—
MPR121	SDA	ESP32-S3	GPIO8
MPR121	SCL	ESP32-S3	GPIO9
MPR121	ADDR	GND rail	—
MPR121	ELE0	Touch pad	Head
MPR121	ELE1	Touch pad	Back
MPR121	ELE2	Touch pad	Belly

Module A	Pin A	→ Module B	Pin B
MPR121	ELE3	Touch pad	Left hand
MPR121	ELE4	Touch pad	Right hand

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7. IMU — GY-87 (I2C)

Module A	Pin A	→ Module B	Pin B
GY-87	VCC	3.3V rail	—
GY-87	GND	GND rail	—
GY-87	SDA	ESP32-S3	GPIO8
GY-87	SCL	ESP32-S3	GPIO9

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8. Distance sensors — TCA9548A mux + 2× GY-53 (I2C)

Module A	Pin A	→ Module B	Pin B
TCA9548A	VIN	3.3V rail	—
TCA9548A	GND	GND rail	—
TCA9548A	SDA	ESP32-S3	GPIO8
TCA9548A	SCL	ESP32-S3	GPIO9
TCA9548A	A0, A1, A2	GND rail	—
TCA9548A	RST	3.3V rail	—
GY-53 #1 (front-left)	SDA / SCL	TCA9548A	SD0 / SC0
GY-53 #2 (front-right)	SDA / SCL	TCA9548A	SD1 / SC1
each GY-53	VCC	3.3V rail	—
each GY-53	GND	GND rail	—
each GY-53	PS	GND rail	—

Both sensors look forward (obstacles); mux channels 2–7 stay empty. PS on GND puts the GY-53 in I2C mode: without it, it doesn't answer. No cliff sensors: nothing stops the robot at a table edge.

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9. Motors — TB6612 + 2× N20 encoder motors

Module A	Pin A	→ Module B	Pin B
TB6612	VM	5V rail	—
TB6612	VCC	3.3V rail	—

Module A	Pin A	→ Module B	Pin B
TB6612	GND	GND rail	—
TB6612	STBY	3.3V rail	—
TB6612	PWMA	ESP32-S3	GPIO1
TB6612	AIN1	ESP32-S3	GPIO2
TB6612	AIN2	ESP32-S3	GPIO42
TB6612	PWMB	ESP32-S3	GPIO41
TB6612	BIN1	ESP32-S3	GPIO13
TB6612	BIN2	ESP32-S3	GPIO47
TB6612	AO1	Left motor	M+
TB6612	AO2	Left motor	M−
TB6612	BO1	Right motor	M+
TB6612	BO2	Right motor	M−
Left motor encoder	VCC	3.3V rail	—
Left motor encoder	GND	GND rail	—
Left motor encoder	C1	ESP32-S3	GPIO17
Left motor encoder	C2	ESP32-S3	GPIO48
Right motor encoder	VCC	3.3V rail	—
Right motor encoder	GND	GND rail	—
Right motor encoder	C1	ESP32-S3	GPIO18
Right motor encoder	C2	ESP32-S3	GPIO3

C1 + C2 together give speed **and** direction. If a wheel counts backwards in software, swap that motor's C1/C2 in code (no rewiring).

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10. Servos — 3× Waveshare SC09 (serial bus)

Each servo has 3 wires: **VIN** (red), **GND** (black/brown), **Signal** (the third one). All three servos connect the same way: same-colour wires joined together.

The servo has **one** Signal wire (it listens and replies on it). The ESP has **two** pins: GPIO20 talks (TX), GPIO19 listens (RX). A 1 kΩ resistor lets both share the one wire:

```
GPIO20 (TX) ──[ 1 kΩ ]──┐
                        ├──── Signal of servo 1 + 2 + 3      ← "SIGNAL point"
GPIO19 (RX) ────────────┘
```

```
5V rail     ──────────────────── VIN of servo 1 + 2 + 3
GND rail    ──────────────────── GND of servo 1 + 2 + 3
```

Module A	Pin A	→ Module B	Pin B	Note
ESP32-S3	GPIO20 (TX)	1 kΩ resistor	either leg	a resistor has no direction
1 kΩ resistor	other leg	SIGNAL point	—	
ESP32-S3	GPIO19 (RX)	SIGNAL point	—	plain wire, no resistor
Servo 1, 2, 3	Signal	SIGNAL point	—	
Servo 1, 2, 3	VIN (red)	5V rail	—	**5V only.** SC09 is rated 4–6 V; 7.4V can destroy it
Servo 1, 2, 3	GND	GND rail	—	

SIGNAL point = one spot where 5 wires meet (resistor leg, GPIO19, three servo Signal wires): a breadboard row, or a solder joint. 1 kΩ colour bands: brown-black-red (4-band) or brown-black-black-brown (5-band).

GPIO19/20 are the board's native USB pins, so that USB port is out of use: **never plug a cable into it**. GPIO43/44 stay free for the board's COM port (its USB-serial chip), which does flashing and the console. (The servo bus used to be on 43/44, but the USB-serial chip drives GPIO44, so the COM port and the servos blocked each other.)

Set servo IDs once before assembly: **1 = head, 2 = left arm, 3 = right arm**.

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11. ESP32-S3 pin summary (quick check)

GPIO	Goes to
1	TB6612 PWMA
2	TB6612 AIN1
3	Right encoder C2
4	Both mics BCLK
5	Both mics LRCL
6	Both mics DOUT
7	MAX98357 DIN
8	SDA — MPR121, GY-87, TCA9548A
9	SCL — MPR121, GY-87, TCA9548A

GPIO	Goes to
10	Display CS
11	Display MOSI
12	Display SCK
13	TB6612 BIN1
14	Display DC
15	MAX98357 BCLK
16	MAX98357 LRC
17	Left encoder C1
18	Right encoder C1
19	Servo bus (direct) — native USB D−
20	Servo bus (via 1 kΩ) — native USB D+
21	Display RST
41	TB6612 PWMB
42	TB6612 AIN2
43, 44	**Leave free** — the board's COM port (USB-serial chip): flashing + console
45	Display EN (backlight)
47	TB6612 BIN2
48	Left encoder C2
38, 39, 40	**Leave free** — used by built-in SD slot on boards that have one
26–37	**Never use** — flash/PSRAM, board won't boot

Cautions

1. Set XY-3606 to **5.0 V with nothing connected**, then wire the rest.
2. **All GNDs common.**
3. Keep motor/servo power wires away from mic/I2C wires.